

Homework 3

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Set up

```
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization
library(readxl) # reading .xlsx files
library(ggeffects)
# storing kelp data as object "kelp"
kelp <- read.csv(here("data", "temp-kelp.csv"))
# creating new object that fits a linear model to the data
kelp_model <- lm(
  kelp_elong ~ temp_c,
  data = kelp)
# storing and cleaning personal data as object "personal_clean"
personal_clean <- read_csv(
  here("data", "ENVS 193DS- Personal Data - Sheet1.csv"),
  # skipping the first 2 messed up rows
  skip = 2) |>
  # dropping the first empty column
  select(-1) |>
  # cleaning the names
  clean_names() |>
  # converting sleep to a graphable value
  mutate(
    sleep_hours =
      hour(hms(amount_of_sleep)) + (minute(hms(amount_of_sleep)) / 60),
    # converting the columns of "date", "caffeine", and "satisfaction" to numbers and dates
    date = as.Date(date, format = "%m/%d/%Y"),
    caffeine = as.numeric(caffeine),
    satisfaction = as.numeric(satisfaction))
```

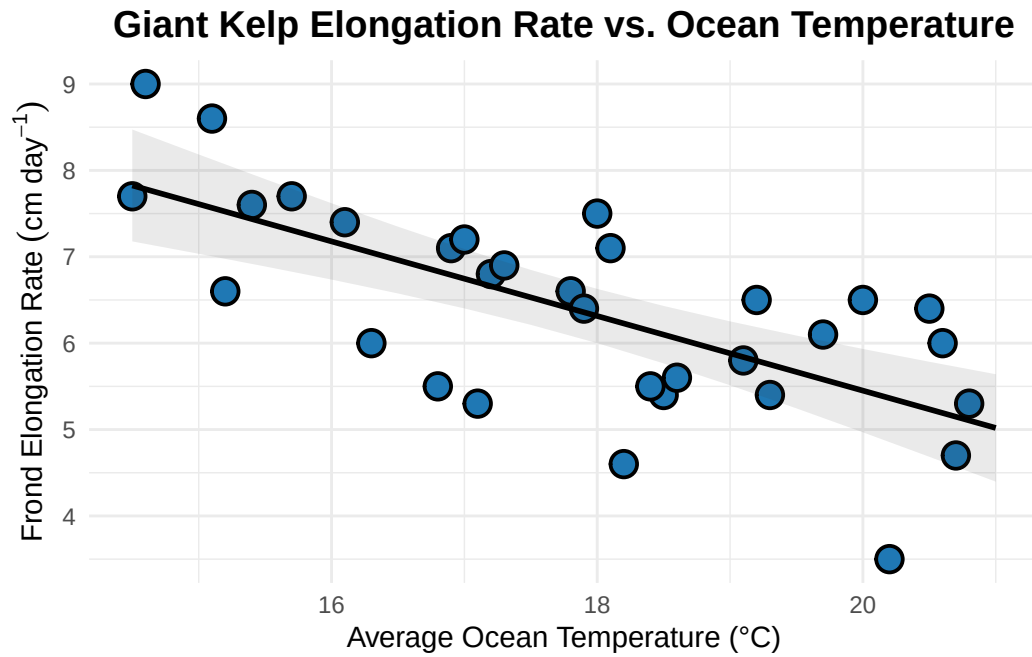
Problem 1

1a.

The appropriate tests would be the Pearson's correlation and linear regression to determine the strength of the relationship between giant kelp frond elongation rate and temperature. Pearson's correlation measures the strength and the direction between the two variables. Linear regression aims to describe a linear relationship between the variables while being able to describe strength as well.

1b.

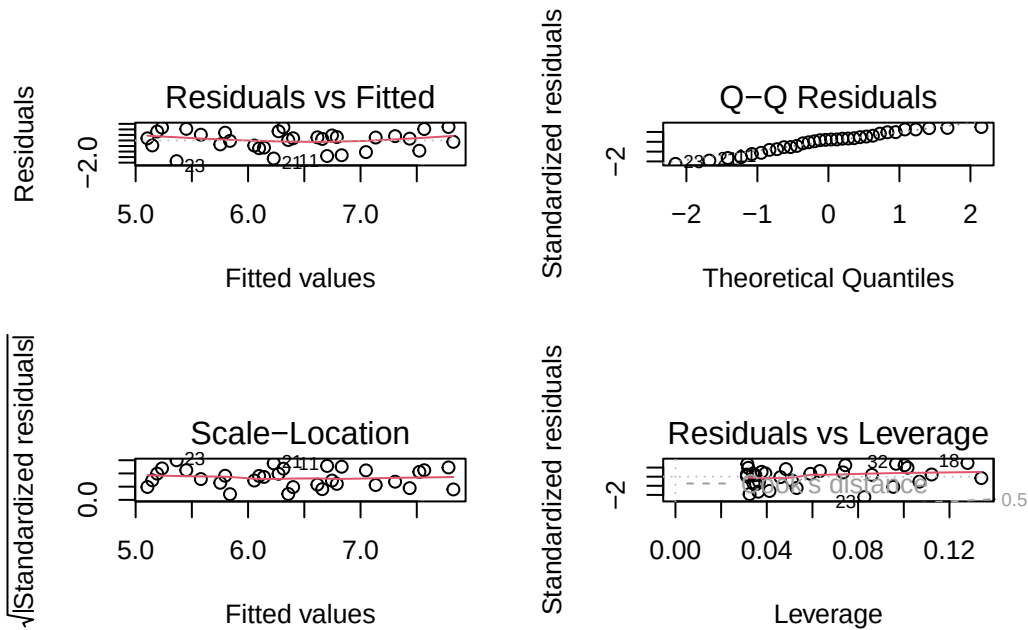
```
# creating new object called kelp_preds
kelp_preds <- ggpredict(
  kelp_model,
  terms = "temp_c")
# base layer: ggplot with kelp data
ggplot(data = kelp,
  aes (x = temp_c, # x-axis is sea temp
        y = kelp_elong)) + # y-axis is kelp elongation rate
# first layer: geom_point to show underlying data
geom_point(fill = "#1f78b4", # changing the color and size
  size = 4,
  stroke = 1,
  shape = 21) +
# second layer: ribbon representing confidence interval
# using predictions data frame
geom_ribbon(data = kelp_preds,
  aes(x = x,
        y = predicted,
        ymin = conf.low,
        ymax = conf.high),
  alpha = 0.1) +
# third layer: line representing model predictions
geom_line(data = kelp_preds,
  aes(x = x,
        y = predicted),
  linewidth = 1)+
labs ( x= "Average Ocean Temperature (°C)",
  y = bquote(Frond~Elongation~Rate~(cm~day^-1)),
  title = "Giant Kelp Elongation Rate vs. Ocean Temperature") +
# changing the theme
theme_minimal() +
# customizing the look of the titles!
theme(plot.title = element_text(face = "bold",
  size = 14,
  hjust = 0.5),
  axis.title = element_text(face = "plain",
  size = 11))
```



1c.

Checking assumptions

```
# base R residuals  
par(mfrow = c(2,2))  
plot(kelp_model)
```

When thinking about the relationship between ocean temperature and kelp elongation, it would make sense for there to be a linear relationship. We can see this in the residuals vs. fitted plot and the plotted data. We can assume independent errors due to the independent observations and data collection methods. The residual vs. fitted and scale location plots both show us the homoscedasticity of errors. The plots show that the data is homoscedastic because of the decently straight trend line. Looking at the QQ plot seen in residual plots, the data seems to be normally distributed. It roughly follows the straight line with variations that are normal to see in real data.

I think I did ok checking the assumptions. It's hard because making the decision is subjective, but generally most math isn't, and making the "wrong" choice can have big impacts. The QQ plot was hardest to assess due to the slight curve seen in it.

Running test

```
# showing the output of linear regression
summary(kelp_model)
```

Call:

```
lm(formula = kelp_elong ~ temp_c, data = kelp)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.8643	-0.5017	0.1790	0.5659	1.2177

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.08645	1.49219	9.440	1.72e-10 ***
temp_c	-0.43179	0.08321	-5.189	1.37e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom

Multiple R-squared: 0.473, Adjusted R-squared: 0.4554

F-statistic: 26.93 on 1 and 30 DF, p-value: 1.366e-05

1d.

To evaluate the strength of the relationship between temperature and giant kelp frond elongation rate, I used an F test from a linear regression model. I used the linear regression model because the data met the assumptions for the model and the goal was to test the strength of the relationship of the variables. A linear model is great for testing strength and relationship. I used the f-test because it's traditional to get the output for an F-test when creating a linear model.

We reject the null hypothesis, so the elongation rate of giant kelp decreased as the average ocean temperature increased (linear regression, $F(30,1) = 26.93$, $R^2 = 0.473$, $p < 0.001$, $\alpha = 0.05$)

Based on Model predictions, for each unit 1 increase in ocean temperature, we expect an decrease of 0.43 ± 0.08 (SE) cm day⁻¹.

1e.

This test used the data collected from kelp in varying ocean temperatures to understand how increasing ocean temperature can affect giant kelp growth. From performing a F test for significance and a linear regression model for strength of the relationship, we were able to understand that when average ocean temp increases, we see a decrease in giant kelp elongation. This means that for giant kelp to be at their healthiest, ocean temperatures cannot continue to increase.

1f.

```
# creating new object for pearson's test from kelp data and running pearson's test
kelp_corr <- cor.test(kelp$temp_c,
                     kelp$kelp_elong,
                     method = "pearson")
# displaying test output
kelp_corr
```

Pearson's product-moment correlation

```
data: kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
      cor
-0.6877489
```

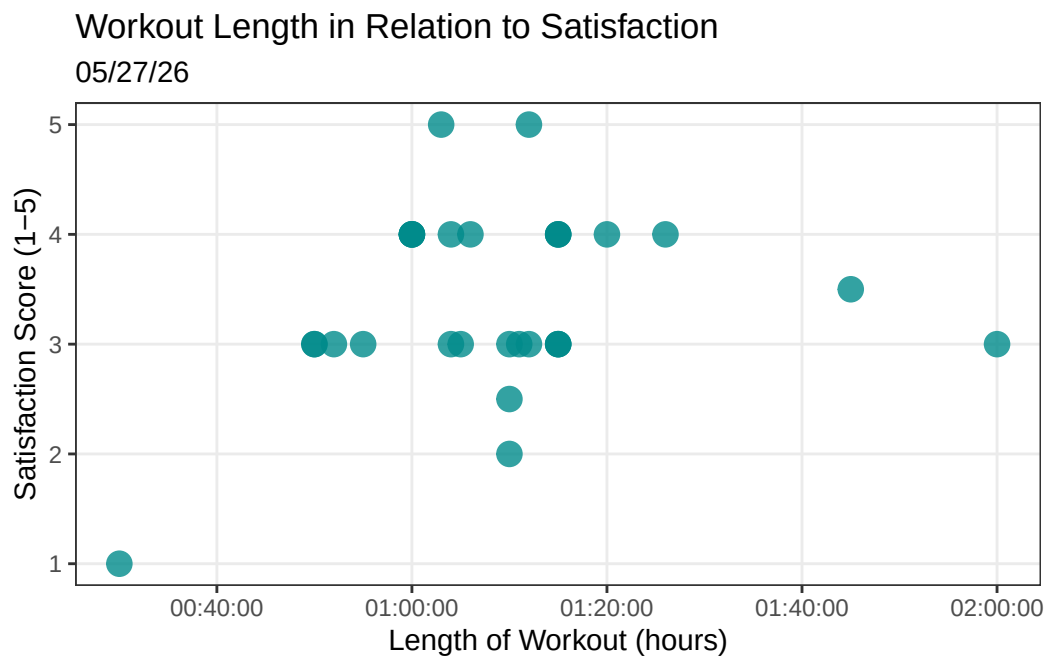
Both tests would have lead to the exact same decision of rejecting the null hypothesis because they have identical t-values of -5.189, p-values of 1.366e-05, and the same significance level of 0.05. They also come to the same conclusion for correlation and strength because of they have to same regression coefficient (-0.688), which is equal to the R squared value of 0.473 when squared. They both show a strong negative correlation between increase in temp and kelp elongation.

Problem 2

2a.

```
# base layer: ggplot
ggplot(data = personal_clean,
       # x-axis is length of wo, y-axis is satisfaction
       aes(x = length_of_wo,
           y = satisfaction)) +
# first layer: point, points with a custom color
```

```
geom_point(color = "cyan4", size = 4, alpha = 0.8) +
# clean theme
theme_bw() +
# removing the excess grid marks
theme(panel.grid.minor = element_blank())+
# labels and title
labs(
  x = "Length of Workout (hours)",
  y = "Satisfaction Score (1-5)",
  title = "Workout Length in Relation to Satisfaction",
  subtitle = "05/27/26")
```



2b.

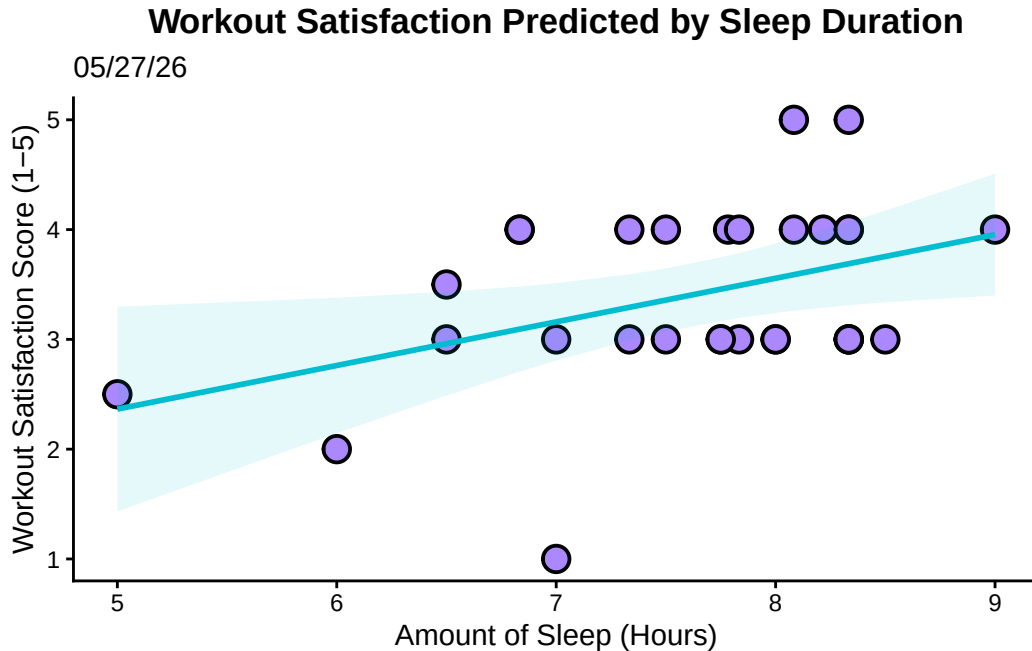
Figure 2. Length of workout doesn't influence satisfaction. Data collected from workouts taken place in the time period 4-20-2026 to 5-27-2026 (Data accessed 05-28-2026). Teal points represent individual observations of length of workout (hours) (total n = 30) plotted against satisfaction with workout (1-5).

```
workout_model <- lm(satisfaction ~ sleep_hours, data = personal_clean)
personal_preds <- ggpredict(workout_model,
```

```

                                terms = "sleep_hours")
#base layer: ggplot
ggplot() +
  # second layer: geom_point to show underlying data with personal_clean data
  geom_point( data = personal_clean,
              aes(x = sleep_hours,
                  y = satisfaction),
              fill = "#ac88ff",
              size = 4,
              stroke = 1,
              shape = 21) +
  # third layer: geom_ribbon to show CI
  # using predictions data frame
  geom_ribbon(data = personal_preds,
             aes(x = x,
                 y = predicted,
                 ymin = conf.low,
                 ymax = conf.high),
             fill = "#00bdd0",
             alpha = 0.1) +
  # third layer: line representing model predictions
  geom_line(data = personal_preds,
            aes(x = x,
                y = predicted),
            color = "#00bdd0",
            linewidth = 1) +
  # adding axes titles and titles!
  labs(
    x = "Amount of Sleep (Hours)",
    y = "Workout Satisfaction Score (1-5)",
    title = "Workout Satisfaction Predicted by Sleep Duration",
    subtitle = "05/27/26") +
  # changing theme
  theme_classic() +
  # removing grid lines, bolding title, and changing axes text
  theme(
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    plot.title = element_text(face = "bold", size = 13, hjust = 0.5),
    axis.title = element_text(face = "plain", size = 11))

```



2b.

Figure 3. Less sleep is associated with lower workout satisfaction. Data collected from workouts taken place in the time period 4-20-2026 to 5-27-2026 (Data accessed 05-28-2026). Purple points represent individual observations of amount fo sleep (hours) (total $n = 30$) plotted against workout satisfaction score (1-5). Teal, thin line represent the linear regression model prediction for workout satisfaction given amount of sleep. Thick, transparent, teal stripe represents 95% confidence interval of linear regression model.

Problem 3

3a.

An affective visualization for my personal data could be done in a variety of ways. My idea is to use circles to represent each workout. The size of the circle is based on sleep, the color is based on satisfaction, the type of workout would be some pattern or symbol on the circle, and a handful of other features for the other variables. This would work for my data because it has the ability to have lots of different features, and my data collection has many different variables.

3b.

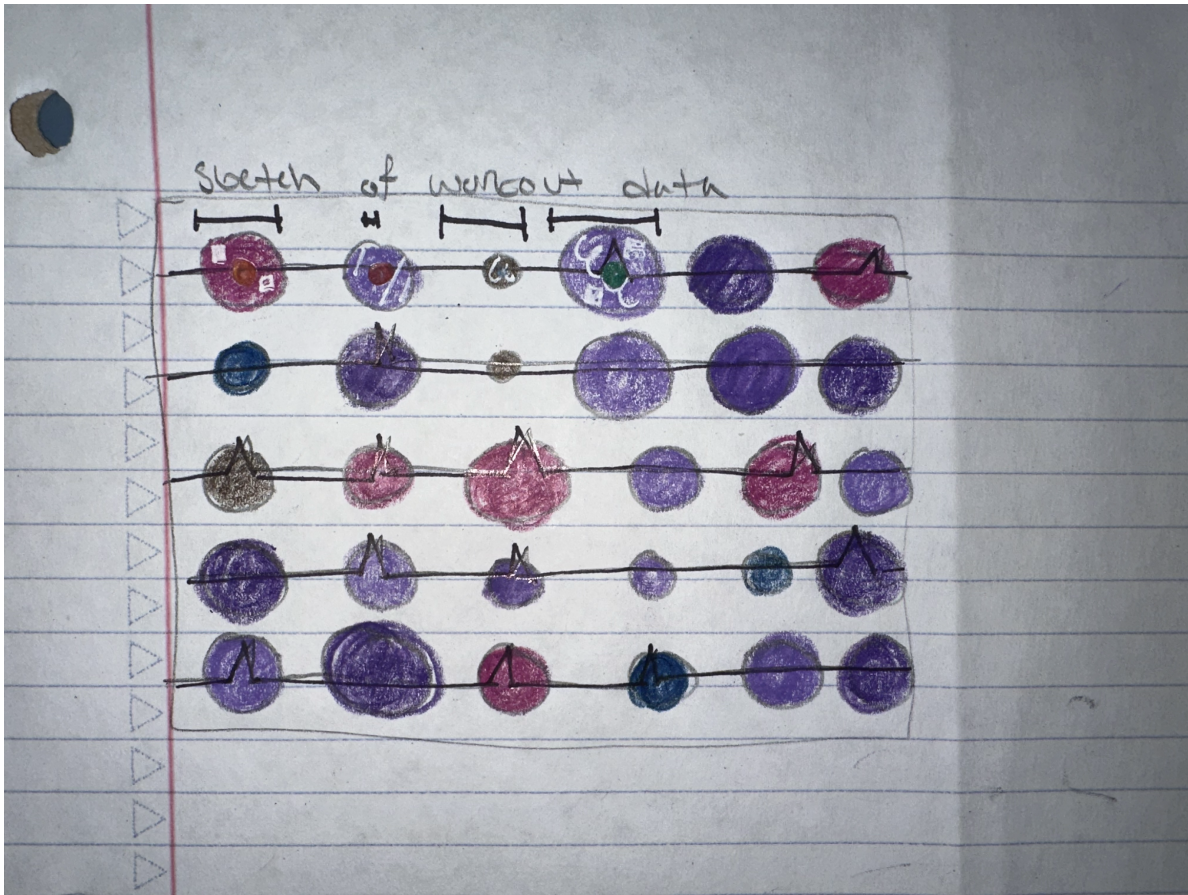


Figure 1: Figure

3c.

3d.

My piece is showing the stats that affect my workouts (amount of sleep, type of workout, caffeine intake, etc.) and the satisfaction I feel after the workout.

I found Lorraine Woodruff-Long visualization on global warming to be the most inspiring. I like the idea of using a shape, levels of color, and even size to demonstrate the data.

I will be doing a mixed media piece with watercolors, colored pencils, and pens to make my piece. Using a variety of materials allows me to showcase more variables.

I created this piece by brainstorming how to show many variables that was in my artistic realm and used inspiration from Lorraine Woodruff-Long. From their I decided to sketch on my paper the region for each circle, make a color palette, and a potential key.

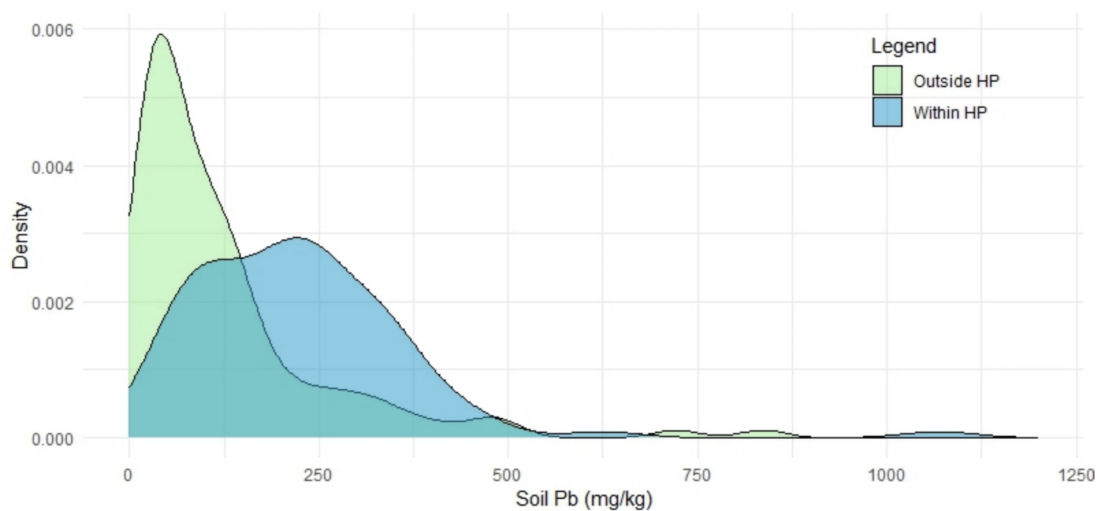
3e.

[Slides](#)

Problem 4

4a.

The test for this paper is Mann-Whitney U. This test addresses the main question of “if the lead contamination in neighborhoods in Huntington Park are higher than other places in LA?” by showing a potential significant difference using a p-value ($p < 0.01$). If there was a significant difference seen, that means that the lead contamination in Huntington Park would be higher than the surrounding areas. This test works for this because they’re independent and not normally distributed.



4b.

The author could have been more clear with their visual statistics. The x- and y- axes were chosen appropriately, but they don’t show any underlying data, means or SE. I think it was wise to fill in the area under the lines, considering this plot is looking at the concentration

density, but it lacks any other data visualization component. I don't feel like it's misleading or under representative per-say, but it could have been a little more descriptive.

4c.

The figure from part a does not have very much visual color. The legend is small, there isn't excess graph features, and the grid lines are light in color. The data:ink ratio is intermediate. Having filled the area under the curve, increases the ratio, but the colors are transparent. Considering the information they're communicating (lead contamination density in soil), it makes sense to fill the area under the curve.

4d.

This paper includes multiple figures, so there isn't much I would recommend changing. I think that showcasing the underlying data could be helpful. We've been taught in this class to do that because it's important to see where exactly the figure came from with the individual observations. I also think that they could have separated the plot into two. Displaying the two plots (one with concentration density outside HP, the other with concentration density within HP) side by side would have been effective as well, and then there wouldn't be any overlap in data (which could confuse some viewers at first glance).